Introduction to Data Structures

Objective:

By the end of this lesson, students will:

1. *Understand what data structures are and why they're important.*
2. *Learn how data structures differ from basic data types.*
3. *Identify and categorize different types of data structures.*

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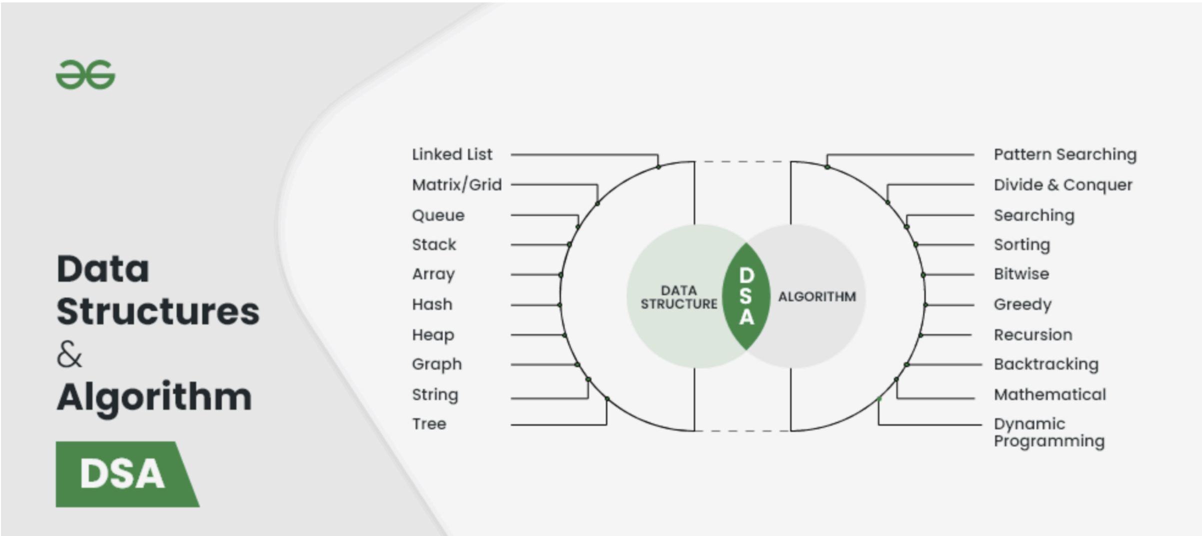
1. Data Structures



1.1. What Are Data Structures?

- **Definition:**
  - **Data structures** are like containers for storing data. Just as different types of containers *(boxes, jars, bags)* are used to store various items efficiently, data structures help organize and manage data in a computer efficiently.

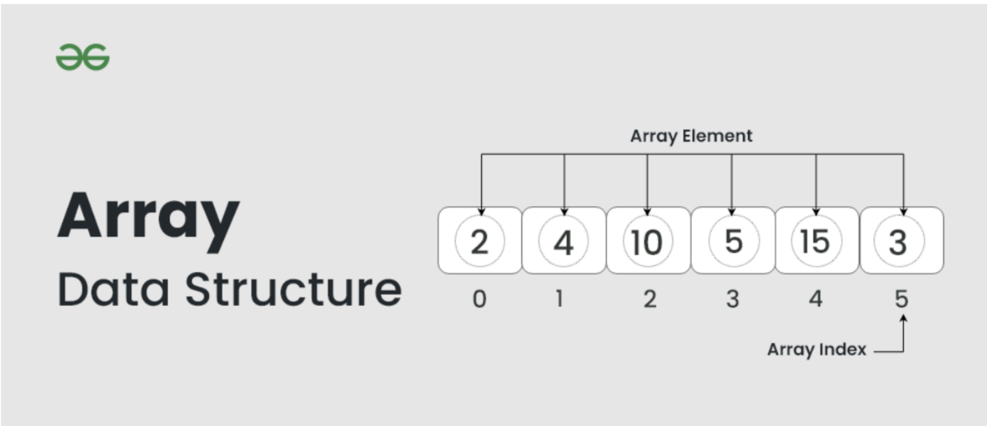
- **Why Data Structures is Important:**
  - Good data structures help computers perform tasks more quickly and use less memory. Imagine trying to find a book in a messy room versus an organized library; data structures help computers keep data organized and easy to find.

1.2. Basic Operations on Data Structures

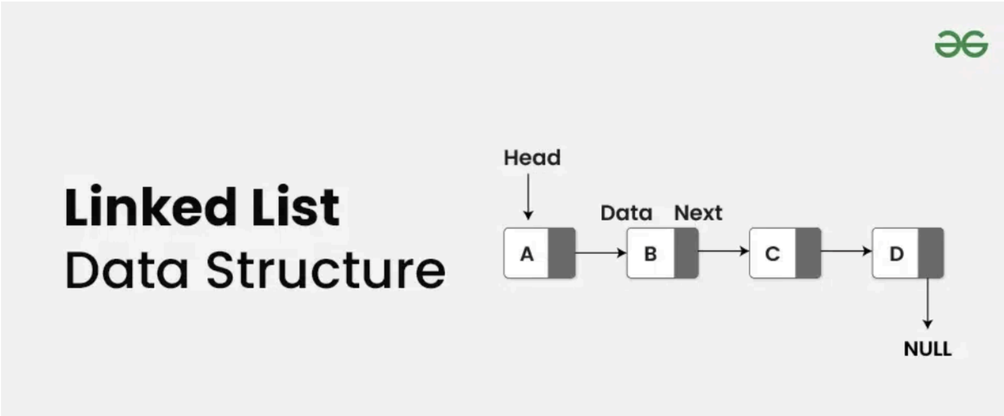
- **Operations:**
  - **Adding Data:** Like putting a new book on a shelf *(Create | Insert).*
  - **Removing Data:** Like taking a book off a shelf*(Delete).*
  - **Accessing Data:** Like opening a book to read it*(Search | Read).*
  - **Updating Data:** Like rewriting or adding notes in a book*(Update).*
- **Use simple examples:**
  - Adding items to a shopping list.
  - Removing items from the list.
  - Checking what's on the list.
  - Update the items on the list.

1.3. Types of Data Structures

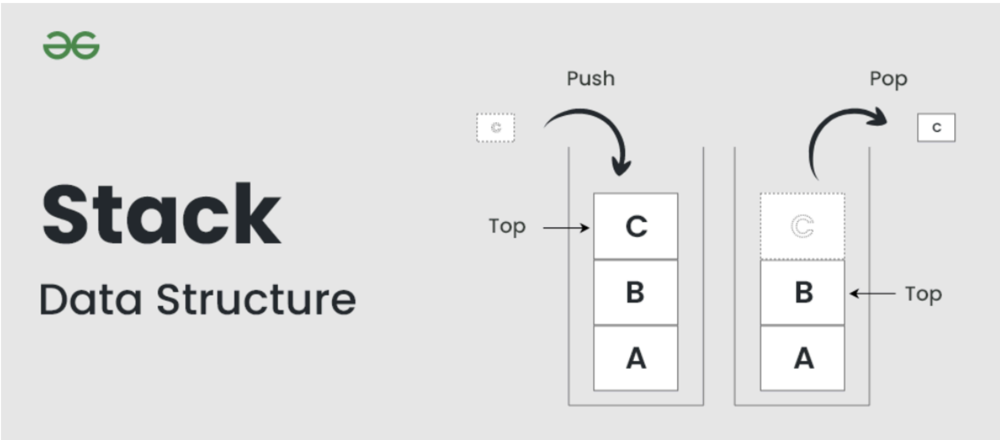
- **Linear Data Structures:**
  - **Arrays:** Like a row of boxes where each box has a number.



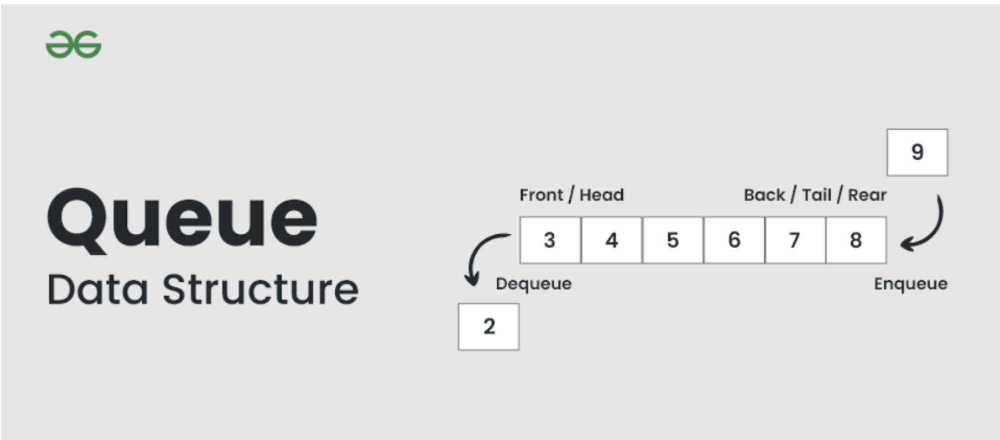
○ **Linked Lists:** Like a chain of linked rings, where each ring points to the next one.



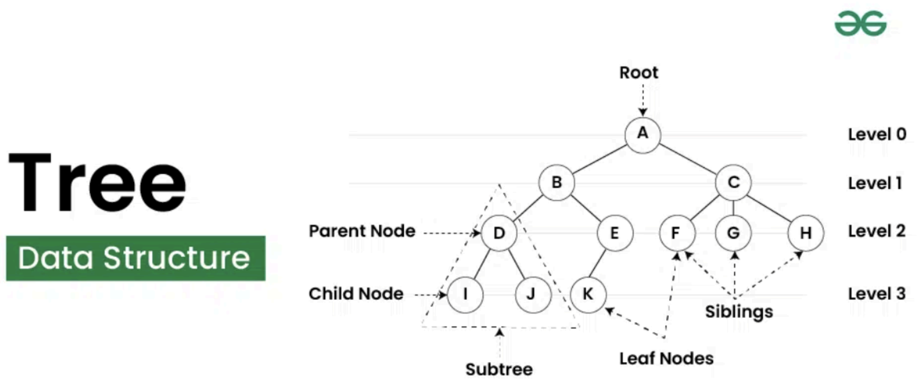
○ **Stacks:** Like a stack of plates where you add and remove from the top.



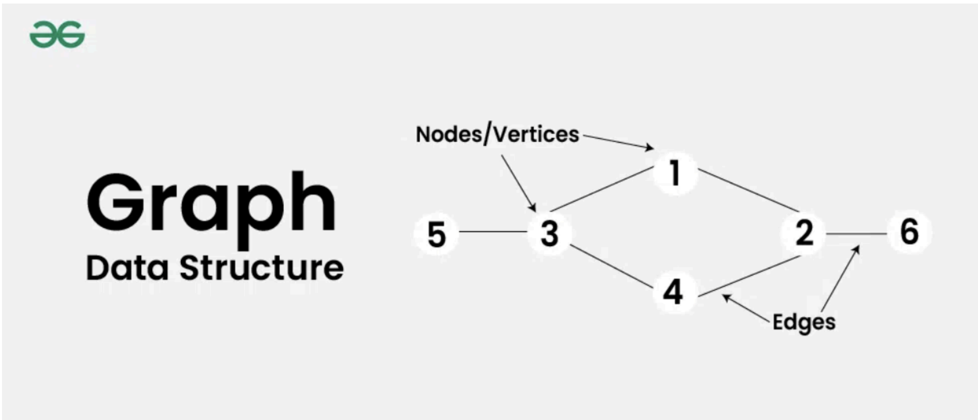
○ **Queues:** Like a line at a checkout where the first person in line is the first one served.



- **Non-Linear Data Structures:**
    - **Trees:** Like an upside-down family tree showing relationships between items.



    - **Graphs:** Like a map showing various points (cities) and connections (roads) between them.



- **Abstract Data Types (ADTs):**
    - **Sets:** Like a collection of unique items, no duplicates allowed.
    - **Maps (Dictionaries):** Like a real-world dictionary where you look up words (keys) to find their definitions (values).

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2. Data Structures vs. Data Types

2.1. Data Types

- **Definition:**
    - **Data types** are basic kinds of data we use, like numbers, characters, or true/false values. They tell the computer what kind of data we're working with.
- **Examples:**
    - **Primitive Data Types:**
        - **int:** Whole numbers, like **1, 2, 3.**
        - **float:** Numbers with decimals, like **3.14.**
        - **char:** Single characters, like **'A'** or **'b'.**
- **User-Defined Data Types:**
    - **Structures:** Custom ways to group different types of data, like a form with multiple fields **(name, age, address)**.

2.2. Differences Between Data Types and Data Structures

- **Data Types:**
    - Think of data types as the ingredients in a recipe—each ingredient is a basic element.
- **Data Structures:**
    - Data structures are like the recipe itself—they organize these ingredients into a dish or meal, allowing you to manage them in a specific way.
- **Example:**
    - A single number (data type) vs. a **list** of numbers (data structure).

3. Classification of Data Structures

3.1. Linear Data Structures

- **Arrays:**
  - **Definition:** A simple list where each item is stored in a specific order.
  - **Example:** An array of numbers like **[1, 2, 3, 4, 5].**
- **Linked Lists:**
  - **Definition:** A list where each item points to the next one in line.
  - **Example:** Think of a chain where each link has a tag pointing to the next link.
- **Stacks:**
  - **Definition:** A collection where the last item added is the first one to be removed (Last In, First Out - LIFO).
  - **Example:** A stack of plates—you add and remove from the top.
- **Queues:**
  - **Definition:** A collection where the first item added is the first one to be removed (First In, First Out - FIFO).
  - **Example:** A queue at a ticket counter where the person who arrives first gets served first.

3.2. Non-Linear Data Structures

- **Trees:**
  - **Definition:** A structure that resembles an upside-down tree, with a root and branches showing how items are related.
  - **Example:** A family tree showing family relationships.
- **Graphs:**
  - **Definition:** A network of points connected by lines, showing how items are linked.
  - **Example:** A map with cities (points) connected by roads (lines).

3.3. Abstract Data Types

- **Sets:**
  - **Definition:** A collection of unique items with no duplicates.
  - **Example:** A set of unique student IDs in a class.
- **Maps (Dictionaries):**
  - **Definition:** A collection of key-value pairs where each key is unique and maps to a specific value.
  - **Example:** A dictionary where each word (key) has a definition (value).
